RUBBER STAMP: SUPPLEMENT

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This supplement accompanies *Rubber Stamp: Gemma 4 Scores Its Own Failing Patches as Sure to Pass* (`rubber-stamp.pdf`). Terms, patch sets and measures are defined there (Section 2). Every number traces to a file in `scripts/derived/` of the repository, https://github.com/Vein05/gemma4-paper-track.

A SETUP DETAILS AND JUDGE PROMPT

Issues. The held-out issues are the first 200 of a fixed random order (seed 0) of SWE-bench Verified issues outside the Mini issues for which at least one of the five other writers has a usable patch (447 issues). Five pilot issues drawn first were left out of the sample. A usable patch is non-empty after removing test files and at most 30,000 characters long.

Agent. The 4-bit competition checkpoint `gemma-4-31B-it-qat-w4a16-ct` (revision `52f3f65b`) ran in mini-SWE-agent v1.7.0 with its SWE-bench configuration unchanged (250 steps, temperature 0, 60-second command timeout), served by vLLM with a 131,072-token context on one NVIDIA L40S. Docker was replaced by Apptainer sandboxes of the same SWE-bench images; grading used SWE-bench 4.1.0's grading code in the same kind of sandbox.

Judges. Outside judges for a model exclude its own family and, for each patch, the family of the patch's writer. Gemma as judge is the hosted full-precision model; in an earlier one-word check on 173 other patches, it and the 4-bit checkpoint gave the same answer in 331 of 344 calls. The re-ask limit was 8,000 output tokens for our own patch sets and 3,000 for the mixed-harness set; other writers' patches on the Mini issues keep their mixed-harness scores.

Prompt. Every judge received the same system and user messages at temperature 0 (`scripts/selfjudge_matrix.py`, `scripts/selection_test.py`). Braces mark the parts filled in per patch.

```
System: You are an expert Python maintainer. You review a proposed
fix for a reported issue. You cannot run code; judge from the
material given.

User: A user reported the issue below for the repository
{owner/repo}.
<issue>{issue text}</issue>
A coding agent produced the patch below.
<patch>{patch without test files}</patch>
Question: after this patch is applied, will the maintainers' hidden
tests for this issue pass? The hidden tests were written for the
upstream fix. They check the behaviour the issue asks for, and
tests that passed before must still pass. Explain briefly, then end
your reply with a final line that is exactly SCORE: <n>, where n is
your probability from 0 to 100 that the hidden tests pass.
```

B AGENT RUNS AND GRADING

Agent runs. The 200 held-out issues ran as two jobs of 100. In the first, 98 runs submitted a patch, one ran out of the 131,072-token context and one stopped after repeated model-call timeouts; three patches were empty. In the second, one submitted patch was empty. On the Mini issues, 49 runs submitted and one stopped after repeated timeouts. Empty patches and runs that stopped count

as unresolved, as in SWE-bench. One Mini issue (`django__django-11848`) failed to start its sandbox and was run again; its second patch has not been graded, so the Mini analyses leave it out.

Reference-patch check. We graded the official reference patch of every issue we ran (5 pilot, 200 held-out, 50 Mini). 248 of 255 resolve in our environment. The first pass labelled 16 matplotlib and xarray issues as timeouts; logs showed the container build, not the tests, exceeded a 1,200-second limit, and with a 3,600-second build limit all but three resolved. The 7 issues whose reference patch does not resolve were dropped from the held-out set (`astropy-7606`, `matplotlib-20826`, `-23314`, `-26113`, `-26208`, `requests-2931`, `sphinx-9711`); Gemma resolved none of them.

Agreement with official labels. Before using our grading, we graded 30 gpt-oss-120b patches with known official labels, stratified by repository. 26 of 30 matched. The four misses were environment failures (a network-dependent test, Ghostscript file access) and patches that apply only with fuzzy matching on today's rebuilt images. Other writers' patches therefore keep their official labels.

Resolve rates. Gemma resolved 56 and 42 of the two held-out batches of 100, 98 of the 193 kept issues in all, and 21 of the 49 Mini issues with a valid label. On the Mini issues two sphinx patches ran past the 1,800-second test limit; counting them as failing, as SWE-bench does, changes the matched-harness gap from $+0.070$ to $+0.084$ [-0.008, 0.184].

C Re-asks and sensitivity

Table 1: Answers without a parsable score on our own patch sets (1,063 patches per judge), re-asked once with an 8,000-token limit. Most had run out of the 3,000-token limit before the score line.

Judge	Re-asked	Recovered	Left out
Gemma 4 31B	1	0	1
gpt-oss-20b	14	14	0
Qwen3.5-9B	74	73	1
GLM-4.7-Flash	52	42	10
Qwen3-Coder	0	0	0
gpt-oss-120b	5	0	5
GLM-4.5	1	1	0
Kimi-K2	129	117	12
Llama-4-Maverick	0	0	0

Sensitivity to the re-ask (exploratory). Leaving out patches whose answer needed a re-ask, the held-out self-judging gap is $+0.030$ [-0.024, 0.083] (all judges' re-asked patches; 763 patches, 143 of Gemma's), $+0.055$ [0.001, 0.106] (only Kimi-K2's) and $+0.063$ [0.012, 0.115] (only Qwen3.5-9B's). Without the re-asked patches, Gemma's own AUROC rises from 0.622 to 0.670, its leniency on failing patches is $+11.1$ [3.8, 18.2] and its rubber-stamp rate 56%.

Leniency on ranks. On per-judge percentile ranks instead of raw scores, the held-out leniency is $+0.085$ [0.048, 0.122] on failing patches and $+0.024$ [-0.007, 0.055] on passing patches.

D What the deficit is not

We checked four simple explanations on the mixed-harness patches, where Gemma has several attempts per issue.

Patch size. Gemma's patches are much smaller than the other writers' (7.7 against 72.7 added lines on average). Among single-file failing patches, matched by size in four bins, Gemma's leniency on its own patches stays 24.1 points [12.7, 39.9] above its leniency on other writers' patches (Table 3).

Missing reasoning. We rated every Mini patch again with Gemma's thinking switched on (Table 2). Its AUROC on its own patches does not move. The gap shrinks only because thinking makes Gemma less accurate on other writers' patches.

Table 2: Gemma 4 31B with reasoning off and with thinking on, Mini issues, mixed-harness patches.

	Reasoning off	Thinking on
AUROC, own patches	0.636	0.638
AUROC, other writers' patches	0.833	0.795
Own failing patches scored 95 or above	65%	64%
Other writers' failing patches scored 95 or above	22%	35%
Self-judging gap [95%]	$+0.135$ [0.060, 0.210]	$+0.097$ [0.010, 0.192]

Issues it cannot solve. Gemma scores 66% of its failing patches at 95 or above on issues that one of its attempts solved (29 patches) and 67% on issues none solved (51). On other writers' failing patches for the unsolved issues, it is slightly stricter than outside judges (-5.8 points). So "a model cannot verify what it cannot solve" does not fit.

Choosing among attempts. We split the pass/fail pairs of Gemma's own patches into pairs from the same issue and pairs from different issues and computed pair accuracy, with ties counted as half. On 47 same-issue pairs Gemma scores 0.628 against 0.646 for the outside judges (-0.019 [-0.243, 0.148]). On 5,553 different-issue pairs it scores 0.636 against 0.706 (-0.070 [-0.138, -0.006]). Gemma gives exactly 100 to 110 of its 150 own patches, so its scores carry little information about which issue a patch belongs to; breaking those ties perfectly would raise its AUROC from 0.64 to 0.91.

We cannot yet tell whether Gemma recognizes its own patches or shares the mistakes in them. It is not more lenient toward other writers' failing patches that resemble its own wrong fix on the same issue ($+5.6$ [-5.5, 17.4]), but most of those patches share almost no changed lines with Gemma's, so this test is weak.

Table 3: Gemma's leniency on single-file failing patches by number of changed lines, mixed-harness Mini patches. The size-matched difference over these bins (own minus others') is $+24.1$ [12.7, 39.9].

Changed lines	Own failing (n)	Others' failing (n)	Leniency, own	Leniency, others'
1–5	29	19	$+14.2$	-7.1
6–15	32	13	$+11.2$	-17.1
16–40	11	3	$+2.6$	-11.6
41 or more	5	5	$+29.1$	-5.6

E MODELS JUDGING THEIR OWN PATCHES

On the Mini issues, five of the other writers judged their own mini-SWE-agent patches, so the harness is matched for them by construction. None shows a gap (Table 4); pooled over the five, the gap is -0.036 [-0.089, 0.016]. Their leniency on their own failing patches is less clear, because each wrote only 14 to 24 failing patches. The point estimates range from -5.4 to $+17.5$ points, without intervals. Llama-4-Maverick's ($+17.5$) is close to Gemma's, but Maverick also scores other writers' failing patches 95 or above more often than its own (69% against 57%), so it does not single out its own patches the way Gemma does.

Table 4: Models judging their own patches on the Mini issues (all patches from mini-SWE-agent).

Model	AUROC, own patches	AUROC, other families' patches	Self-judging gap [95%]
Qwen3-Coder	0.691	0.786	-0.032 [-0.241, 0.189]
gpt-oss-120b	0.860	0.701	-0.046 [-0.133, 0.049]
GLM-4.5	0.757	0.752	-0.047 [-0.174, 0.116]
Kimi-K2	0.881	0.805	-0.036 [-0.113, 0.036]
Llama-4-Maverick	0.786	0.657	-0.019 [-0.133, 0.101]

F PER-JUDGE RESULTS ON THE HELD-OUT ISSUES

Table 5: Held-out issues, per judge: share of failing patches scored 100 on Gemma's patches and on the other writers' patches; AUROC on Gemma's patches; and, for outside judges, the drop from their AUROC on other writers' patches to their AUROC on Gemma's (the mean drop, 0.055, enters the self-judging gap).

Judge	Gemma's failing at 100	Others' failing at 100	AUROC, Gemma's	Drop
Gemma 4 31B	61.2%	34.6%	0.622	
Llama-4-Maverick	45.9%	38.7%	0.611	0.059
GLM-4.7-Flash	36.5%	24.9%	0.636	0.091
Qwen3.5-9B	21.2%	23.4%	0.685	0.050
GLM-4.5	10.6%	9.2%	0.778	0.053
Kimi-K2	1.2%	1.8%	0.766	0.055
gpt-oss-20b	0%	0%	0.665	0.038
gpt-oss-120b	0%	0%	0.656	0.074
Qwen3-Coder	0%	0%	0.717	0.022

G CHECKS OF GEMMA'S PATCHES

Table 6: AUROC on Gemma's own patches minus Gemma's AUROC on the same patches, with 95% and Bonferroni (eight judges) intervals. Exploratory.

Check	Held-out [95%]	Held-out [Bonferroni]	Mini, mixed / matched
GLM-4.5	$+0.156$ [0.083, 0.228]	[0.054, 0.256]	$+0.082$ / $+0.110$
Kimi-K2	$+0.144$ [0.081, 0.205]	[0.061, 0.230]	$+0.129$ / $+0.037$
Mean of 8 outside judges	$+0.129$ [0.067, 0.194]	[0.039, 0.216]	$+0.130$ / $+0.090$
Qwen3-Coder	$+0.095$ [0.025, 0.163]	[-0.004, 0.187]	$+0.098$ / -0.052
Qwen3.5-9B	$+0.063$ [-0.007, 0.138]	[-0.038, 0.163]	$+0.127$ / $+0.010$
gpt-oss-20b	$+0.043$ [-0.035, 0.128]	[-0.067, 0.156]	$+0.032$ / -0.020
gpt-oss-120b	$+0.034$ [-0.036, 0.106]	[-0.064, 0.131]	$+0.059$ / $+0.117$
GLM-4.7-Flash	$+0.014$ [-0.066, 0.090]	[-0.099, 0.118]	$+0.064$ / -0.105
Llama-4-Maverick	-0.011 [-0.083, 0.060]	[-0.116, 0.092]	-0.032 / -0.055
Gemma + GLM-4.5	$+0.146$ [0.094, 0.197]	[0.074, 0.213]	$+0.083$ / $+0.102$
Gemma + Kimi-K2	$+0.134$ [0.088, 0.181]	[0.070, 0.196]	$+0.114$ / $+0.033$
Gemma + Qwen3-Coder	$+0.101$ [0.053, 0.151]	[0.030, 0.169]	$+0.114$ / $+0.048$
Gemma + Qwen3.5-9B	$+0.081$ [0.032, 0.136]	[0.011, 0.154]	$+0.119$ / $+0.051$

H SUBMITTING WITHOUT REVIEW: ALL RATES

Table 7: Share of submitted patches that pass, by acceptance rate and check. Gemma's own cut (every patch it scores 100) accepts 73.9% of held-out patches, 75.6% of matched Mini patches and 73.3% of mixed Mini patches, with 60.9%, 61.3% and 55.5% passing. Base rates: 52.8%, 48.8%, 46.7%.

Patch set	Check	Top 25%	Top 50%	Top 75%
Held-out (180)	Gemma	60.9%	60.9%	61.0%
	Gemma + Qwen3.5-9B	75.8%	67.5%	61.1%
	Kimi-K2	77.0%	72.4%	63.7%
	GLM-4.5	83.6%	73.5%	63.0%
Mini, matched (41)	Gemma	61.3%	61.3%	61.3%
	Gemma + Qwen3.5-9B	70.0%	69.2%	61.0%
	Kimi-K2	64.5%	66.2%	60.0%
	GLM-4.5	70.6%	71.0%	65.0%
Mini, mixed (150)	Gemma	55.5%	55.5%	54.7%
	Gemma + Qwen3.5-9B	71.2%	64.7%	56.6%
	Kimi-K2	68.8%	67.9%	57.8%
	GLM-4.5	68.3%	62.4%	56.9%

I GEMMA'S OWN SIGNALS

Table 8: Signals that need no second model, on 172 of Gemma's mixed-harness patches with both Gemma scores (planned before computing; `spec/self-check-fix.md`). The planned test was the score plus attempt agreement (mean overlap of changed lines with Gemma's other attempts at the issue); its bar was a gain of at least 0.05 with a lower bound above zero.

Signal	AUROC	Change from Gemma's score [95%]
Gemma's score, reasoning off	0.646	
Gemma's score, thinking on	0.645	-0.001 [-0.056, 0.058]
Mean of the two scores	0.677	$+0.031$ [-0.005, 0.069]
Attempt agreement	0.665	$+0.019$ [-0.119, 0.146]
Score plus attempt agreement (planned)	0.720	$+0.074$ [-0.014, 0.160]
Score relative to its other attempts	0.533	-0.113 [-0.197, -0.025]

J COST

Table 9: API cost as reported by OpenRouter per call. gpt-oss-120b calls were reported at zero cost by their provider. Gemma's 250 agent runs took 49 agent-hours on one NVIDIA L40S of a university cluster, at no charge.

Run	Calls	Cost (USD)
Four judges on Gemma's own attempts (ratings reused)	3,244	0.53
Nine judges, Mini issues, mixed-harness patches	3,292	2.66
Gemma 4 31B with thinking on, Mini issues	401	0.51
Nine judges, held-out and matched-harness patches	9,857	8.66
Total	16,794	12.36